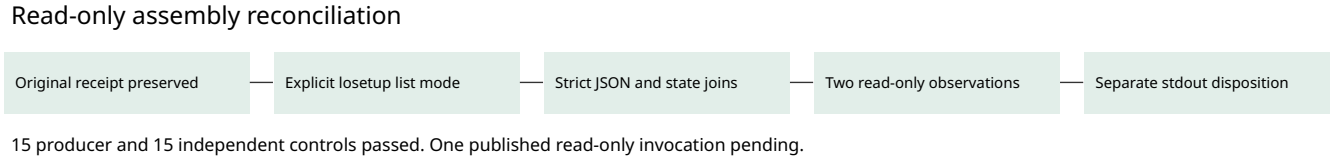
Read-only assembly reconciliation helper

The observed launcher failure comes from parsing empty output as JSON. On installed util-linux 2.37.4, losetup –json –associated DISK returned exit 0 with empty stdout; adding –list returned a JSON object containing an empty loopdevices list. The local manual defines –json for –list output. A separate helper now requests list mode explicitly and rejects empty bytes.



Checks. Fifteen mocked controls pass: valid empty JSON, attached devices, null/wrong/duplicate shapes, stopped and collected unit states, owner and invocation joins, worker completion and explicit teardown booleans. The helper verifies the original source and binding hashes, saved successful completion, root-owned receipt custody and unchanged retained disk custody. It takes two observations of the stopped unit, empty cgroup, associated-device list and entire loop inventory. An unrelated loop conservatively leaves this specific recovery unresolved.

Boundaries. The helper has no mount, unmount, daemon, build, service mutation or qualification path. It emits a new observation on stdout and preserves the original failed launcher receipt. Independent source review accepted the exact helper and replayed all 15 controls. One published bounded read-only invocation by root is the next step. The retained disk hash comes from the worker under unchanged metadata custody; disk contents are not rehashed by this helper. An observation does not authorize qualification.

Validation limits. One producer invocation used a 30-second outer timeout plus 3-second kill reserve, 512 MiB address space and 25-second child timeout. Proposed read-only operation bounds are 30 seconds total, 5 seconds or remaining time per query, with an external 33-second cutoff. No assembly retry or scientific trajectory was run for this packet.